

Satyam Awasthi

Researcher & Software Engineer

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Professional Summary

Computer Science researcher focused on **Autonomous Systems**, **Spatial Perception**, and **Model-Based and Learning-Based Control**. My research explores how perception, predictive modeling, and planning can be integrated with control frameworks to enable robust long-horizon autonomy in robotic systems operating under uncertainty.

Research Interests

Autonomous Systems, Spatial Perception, Model-Based and Learning-Based Control, and Robot Planning, with emphasis on long-horizon decision-making under uncertainty in real-world robotic environments.

Education

University of California, Santa Barbara (UCSB)

M.S. in Computer Science (GPA: 3.91/4.00)

Santa Barbara, CA

Sep 2021 – Jun 2023

Indian Institute of Technology, Kharagpur (IIT Kharagpur)

B.Tech. in Electrical Engineering, Minor in Computer Science (GPA: 8.98/10.0)

Kharagpur, India

Jul 2016 – Jul 2020

Research Experience

Four Eyes Laboratory, University of California, Santa Barbara

Sep 2021 – Jun 2023

Graduate Researcher

Advisor: [Prof. Tobias Höllerer](#), Co-Advisor: [Prof. Michael Beyeler](#)

- Developed **EyeTTS**, an open-source framework for evaluating and calibrating eye tracking during mixed-reality locomotion, resulting in first-author publications at IEEE ISMAR Adjunct and IEEE VR Workshops.
- Designed experimental protocols and software pipelines to quantify gaze-tracking drift during active user motion.
- Curated a multi-device eye-tracking dataset and conducted user studies across diverse locomotion layouts.
- Published first-author poster papers at IEEE VR and IEEE ISMAR Adjunct venues.

Autonomous Ground Vehicle & Swarm Robotics Group, IIT Kharagpur

Jan 2016 – Dec 2018

Undergraduate Researcher

- Contributed to the development of **Eklavya 6.0**, the IIT Kharagpur autonomous ground vehicle that finished **Runner-Up** at the **Intelligent Ground Vehicle Competition (IGVC) 2018**. [Competition Video](#) · [Team AGV](#)
- Developed a hardware-agnostic navigation stack integrating EKF-based localization from LiDAR, IMU, and GPS streams.
- Implemented visual obstacle segmentation using a custom U-Net architecture and hybrid motion planning using A*/Dijkstra and Dynamic Window Approach.
- Designed decentralized multi-agent coordination using relative-pose estimation over an ad-hoc mesh network.

Publications

Peer-Reviewed Conference Papers

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing

Snigdha Das, **Satyam Awasthi**, Abdul Shannar P, Pradipta De, Sandip Chakraborty, and Bivas Mitra.

COMSNETS, 2022. DOI: [10.1109/COMSNETS53615.2022.9668531](https://doi.org/10.1109/COMSNETS53615.2022.9668531)

Workshop & Poster Publications

Eye Tracking Performance in Mobile Mixed Reality

Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, and Tobias Höllerer.

IEEE VRW, 2024. DOI: [10.1109/VRW62533.2024.00321](https://doi.org/10.1109/VRW62533.2024.00321)

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion

Satyam Awasthi, Vivian Ross, Michael Beyeler, and Tobias Höllerer.

ISMAR Adjunct, 2023. DOI: [10.1109/ISMAR-Adjunct60411.2023.00104](https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00104)

Research Artifacts

EyeTTS Calibration Framework

2022–2024

[Calibration Framework](#) · [User Study Framework](#)

- Developed an extensible framework for evaluating and calibrating eye-tracking systems during mixed-reality locomotion.
- Implemented device-agnostic benchmarking pipelines and automated drift-correction procedures.
- Released reproducible software infrastructure supporting eye-tracking research.

EyeTTS Mixed-Reality Eye Tracking Dataset

2022–2024

[Dataset](#) · [User Study Videos](#)

- Curated a multi-device mixed-reality locomotion dataset for calibration and performance evaluation.
- Collected gaze trajectories, calibration measurements, and locomotion metadata across diverse MR layouts.
- Supported experimental validation reported in ISMAR and VRW publications.

Selected Projects

JitterNot

Jan 2022 – Mar 2022

Learning-Based Model Predictive Control for Adaptive Video Streaming (UCSB)

- Developed an LSTM-based predictor for network throughput enabling closed-loop control under non-stationary dynamics.
- Designed a model predictive control (MPC) framework using receding horizon optimization for adaptive decision-making over bandwidth constraints.

IGVC Lane Detection & Navigation

2018

ROS-Based Lane Perception Pipeline for Autonomous Ground Vehicles (IIT Kharagpur)

- Developed a ROS-integrated lane perception pipeline using OpenCV, bird's-eye-view projection, CUDA/GPU-accelerated SLIC-style superpixel segmentation, and parabolic lane fitting for real-time autonomous navigation.
- Rejected obstacle and non-lane regions using contour-based ROI extraction, HOG descriptors, and SVM classification, then published local waypoint goals and segmented top-view lane masks for ROS navigation.

In Motion — HAR using DeepConvLSTM

Nov 2019

Embedded Sensor Inference System for Temporal Activity Recognition (IIT Kharagpur)

- Designed a streaming inference pipeline for high-frequency IMU signals enabling real-time temporal classification.
- Implemented a DeepConvLSTM architecture combining convolutional feature extraction with temporal sequence modeling for activity recognition.

Battleship Solitaire Solver

Apr 2022 – Jun 2022

Solver-Aided Constraint Synthesis for Structured Search Problems (UCSB)

- Formulated Battleship Solitaire as a symbolic constraint-satisfaction problem in Rosette/Racket, encoding grid cells, ship placements, row/column tallies, adjacency rules, and revealed-cell constraints.
- Used Rosette's solver-aided execution to synthesize valid board configurations satisfying all puzzle constraints.

Industry Experience

Intuit Inc

Mountain View, CA · Mar 2024 – Jan 2026

Software Engineer 2

- Built **dual-stage agentic decision pipelines** using tool-schema routing for real-time intent classification and sequential state tracking.
- Developed a decoupled **generative UI engine** leveraging RAG and structured LLMs to optimize deterministic component layouts.

Yahoo Inc

Mountain View, CA · Apr 2023 – Nov 2023

Software Dev Engineer II (IC3)

- Developed **client-side rule engines** for asynchronous receipt payloads, executing deterministic filtering for **real-time decision support**.

Coupang Global LLC

Mountain View, CA · Jun 2022 – Sep 2022

Software Engineering Intern

- Engineered **context-aware heuristics** to prune multi-dimensional search spaces based on active user state vectors.
- Designed **Content-Based Filtering (CBF)** layers using **cosine similarity** to rank high-dimensional product features.

Samsung Research Institute

Noida, India · Sep 2020 – Sep 2021

Software Engineer (CL 2)

- Architected a **continuous on-device perception pipeline** ingesting multi-modal media streams for automated ML classification.
- Designed a **cross-profile data brokering engine** leveraging multi-user storage isolation to route assets from **User 1** to sandboxed **User 0**.

Teaching

University of California, Santa Barbara

Santa Barbara, CA

Graduate Teaching Assistant

Sep 2021 – Jun 2023

- Directed instructional labs, formulated algorithmic programming assignments, and monitored system architecture milestones for senior capstone teams.
- Covered advanced and core tracks: CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).

Technical Skills

Programming: Python, C++, Java, Kotlin, MATLAB

Robotics & Simulation: ROS/ROS2, Gazebo, RViz, Unity

Machine Learning & Vision: PyTorch, OpenCV, CUDA/GPU Acceleration

Methods: State Estimation (EKF), Model Predictive Control (MPC), Motion Planning, Sensor Fusion, Spatial Perception

Developer Tools: Git

Academic Service & Peer Review

Peer Reviewer, UIST 2024

ACM Symposium on User Interface Software and Technology

- Reviewed submissions on interactive systems, spatial tracking, and interface hardware.